11. evaluate $\int_1^1 \sqrt{1+x^4}\,dx = 0$ (一条线积分是0)

let $f(x) = \sqrt{1+x^4}$

$f'(x) = \frac{4x^3}{2\sqrt{1+x^4}}$

when $x = 1$

$f'(x)$

15. evaluate $\int_1^3 e^{x+2}\,dx$

let $f(x) = e^{x+2}$

$f'(x) = e^{x+2} \cdot 1$

when $1 < x < 3$, $f'(x) > 0$, $f(x)$ ↑

∴ when $x = 1$, $f(x)$ has a minimum value of $f(1) = e^3 = 20.09$

$x = 3$. maximum $f(3) = e^5 = 148.41$

Using boundness properties,

$20.09(3-1) \le \int_1^3 e^{x+2} \le 148(3-1)$

$40.18 \le \int_1^3 e^{x+2} \le 296.82$

58. $\int_2^8 f(x)\,dx = 7.3$. $\int_2^4 f(x)\,dx = 5.9 \Rightarrow \int_4^8 f(x)\,dx$

$$\int_4^8 f(x)\,dx = \int_2^8 f(x)\,dx - \int_2^4 f(x)\,dx$$

$$= 7.3 - 5.9 = 1.4$$

$$\int_4^8 dx = \int_4^{} dx + \int_8^{} dx$$

$$= \int_4^6 dx + \int_a^b dx + \int_b^8 dx$$

$$2 \le \int_{-1}^{1} \sqrt{1+x^2}\, dx \le 2\sqrt{2}$$

let $f(x) = \sqrt{1+x^2}$

$$f'(x) = \frac{2x}{2\sqrt{1+x^2}}$$

let $g(x) = 2x$

when $-1 < x < 0$, $g(x) < 0$, $f'(x) < 0$, $f(x)$ ↓

when $0 < x < 1$, $g(x) > 0$, $f'(x) > 0$, $f(x)$ ↑



when $x = 0$, $f(x)$ has a minimum value of $f(0) = 1$

$f(-1) = f(1) = \sqrt{1+1} = \sqrt{2}$ (maximum)

$$1 - (-1) \le \int_{-1}^{1} \sqrt{1+x^2}\, dx \le \sqrt{2}\,[1-(-1)]$$

$$2 \le \int_{-1}^{1} \sqrt{1+x^2}\, dx \le 2\sqrt{2}$$